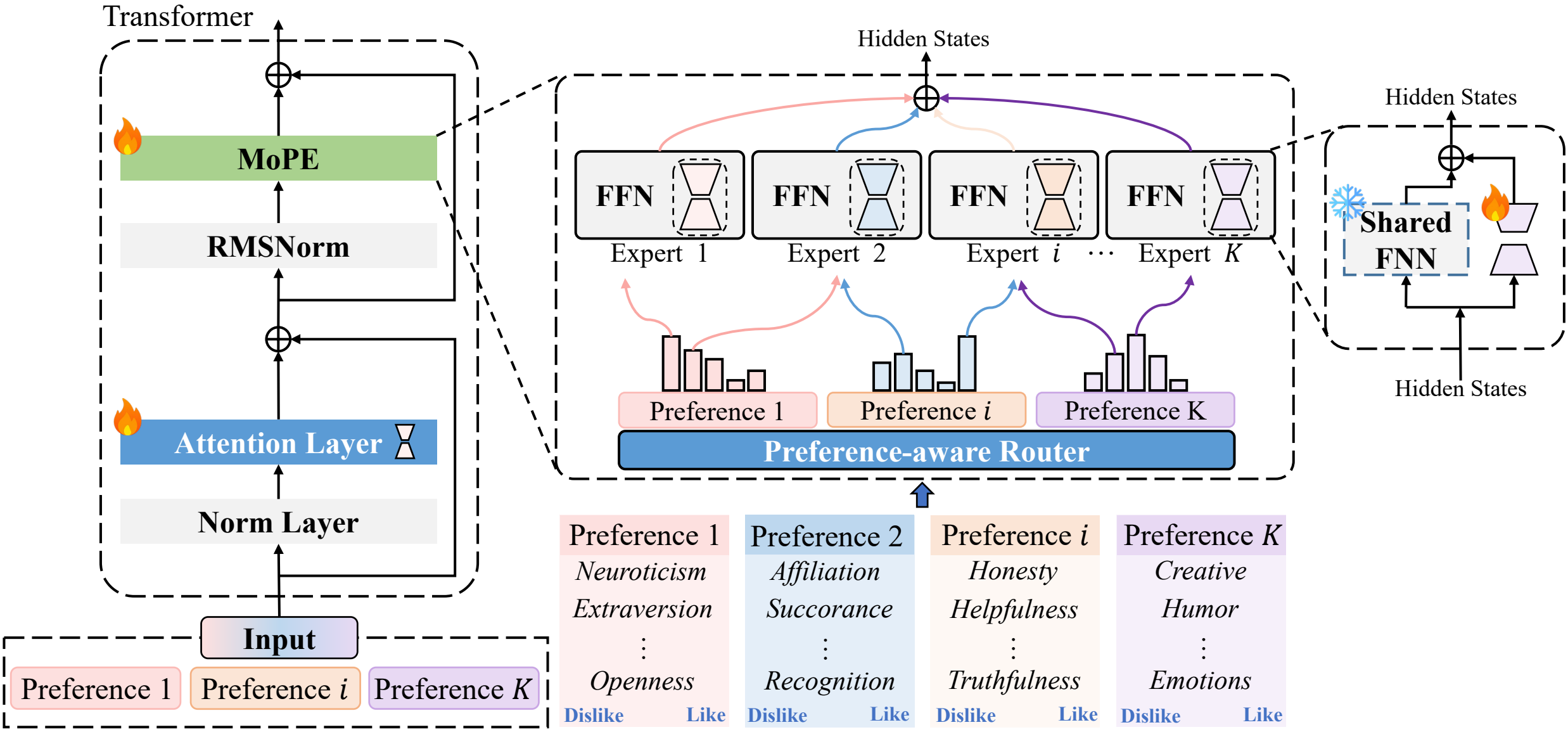
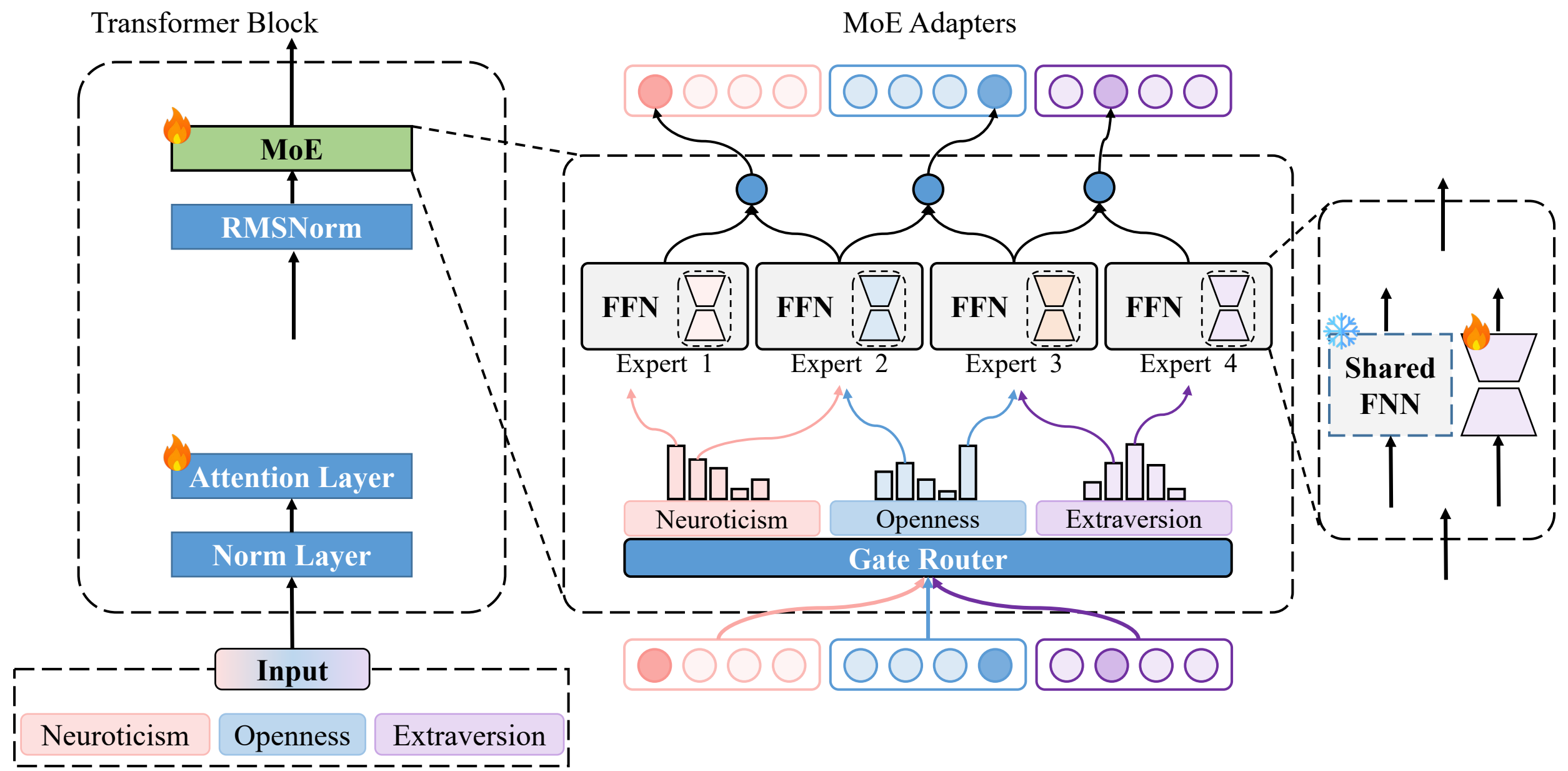
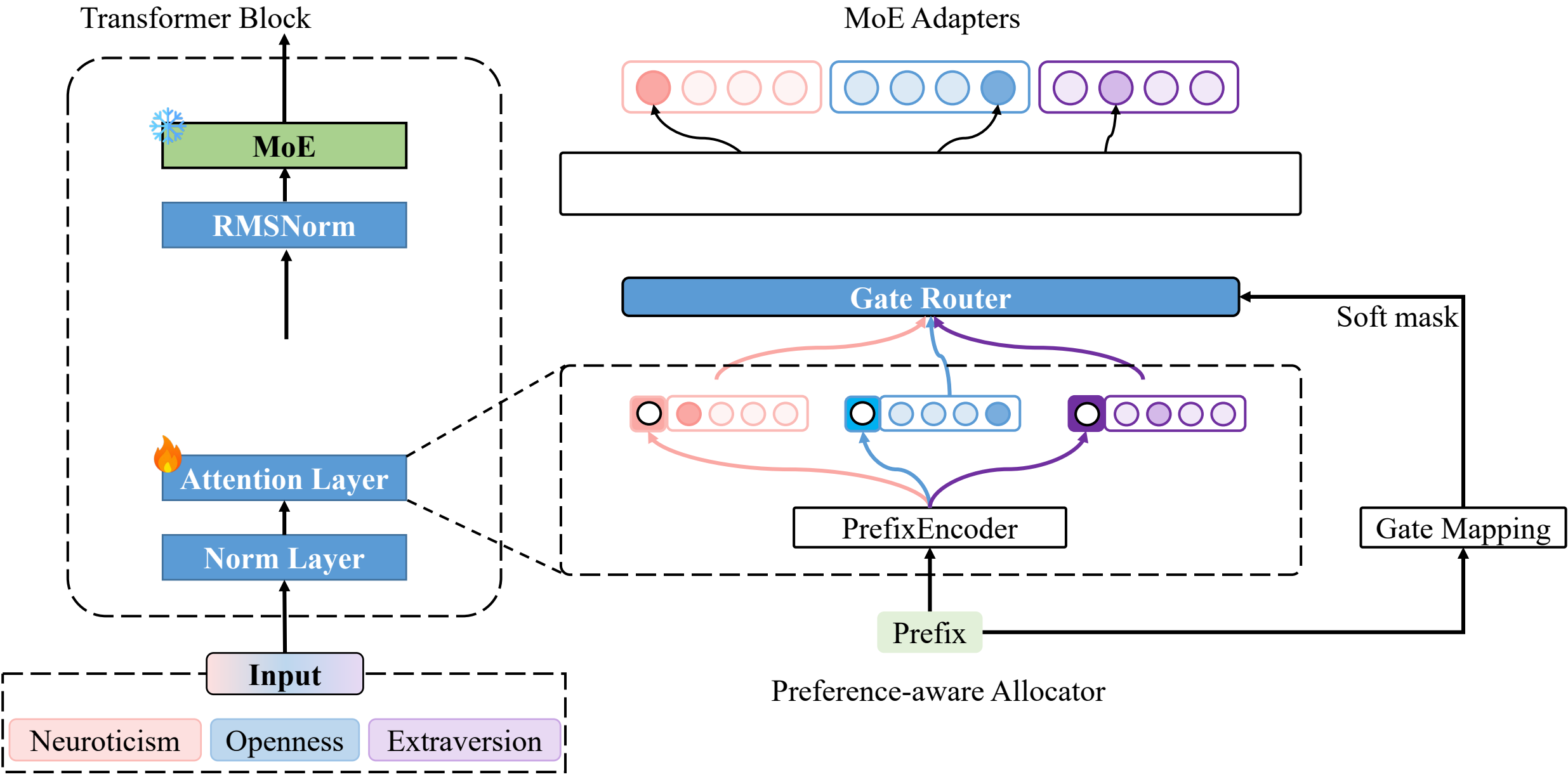




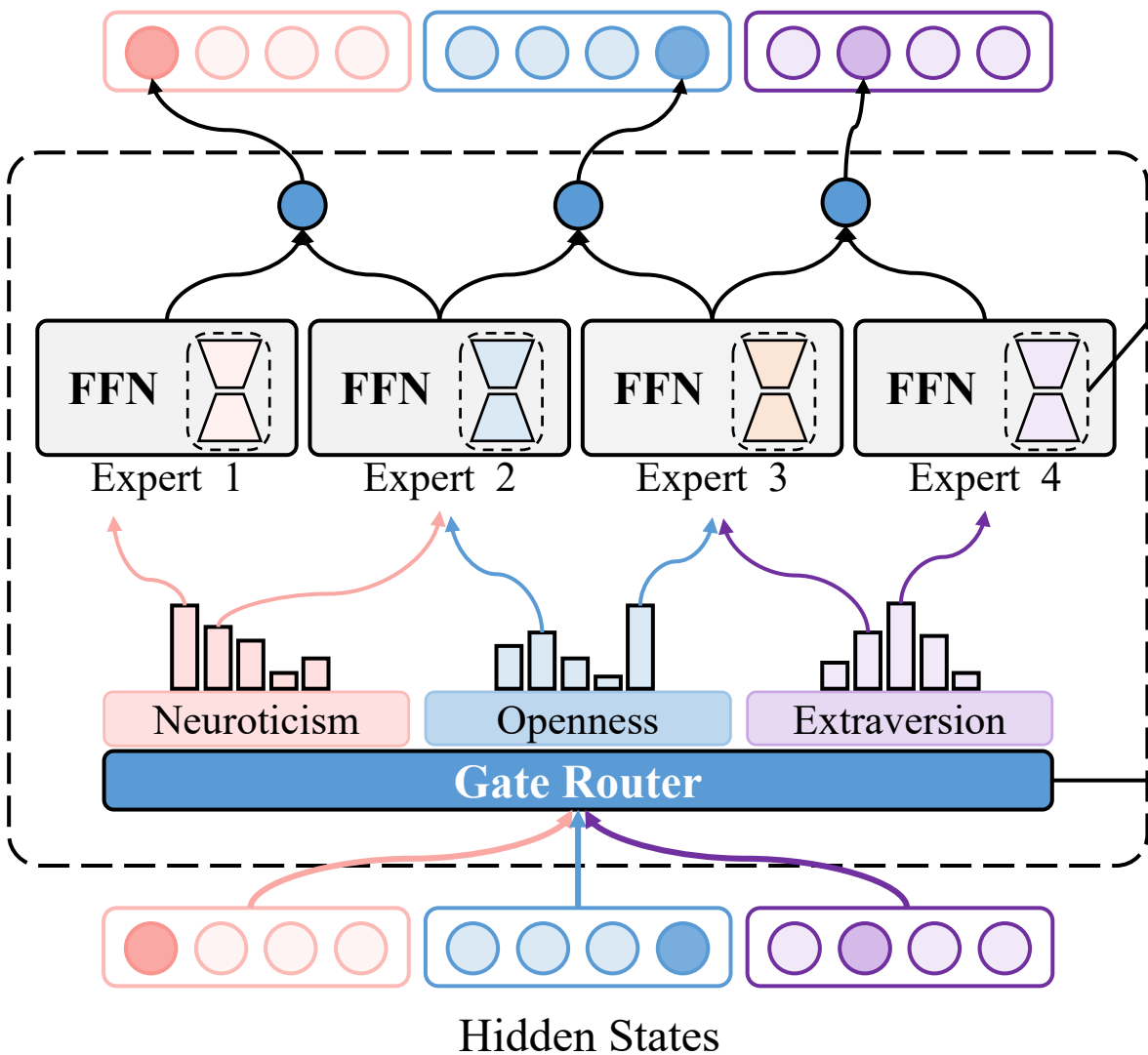
Step 1



Step 2



MoE Adapters



LoRA Expert: FFN解码阶段无法利用prefix tuning在单个Expert内部实现不同方向的解码。因此对FFN的LoRA Experts仅考虑同偏好维度的处理。

- 同一Expert组内由外部prefix embedding引导hidden states来控制，在expert内部不显示感知处理
- 不同Expert组间由contrastive loss来避免耦合

(优势：1. 可解释、可裁剪 2. 更强的动态、组合泛化能力)

Router: 现有MoE的router主要解决负载均衡，导致experts间功能的可解释性是含糊不清的。而由于Transformer是token-level的，因此跳过架构将整个sentence送入特定expert不合理且会损失动态、泛化能力。因此我设计的路由方案是标签软掩码

1 Demographics
("Young", "Older"),
("Female", "Male"),

2 Personality Traits (Big Five)
("High Neuroticism", "Low Neuroticism")
("High Extraversion", "Low Extraversion")
("High Openness", "Low Openness")
("High Agreeableness", "Low Agreeableness")
("High Conscientiousness", "Low Conscientiousness")
("Strong need for Order", "Indifferent to orderliness")
("Strong need for Retention", "Indifferent to retention")
("Strong need for Inviolacy", "Indifferent to inviolacy")
("Strong need for Infavoidance", "Indifferent to Infavoidance")
("Strong need for Counteraction", "Indifferent to Counteraction")
("Strong need for Abasement", "Indifferent to Abasement")
("Strong need for Contrariance", "Indifferent to Contrariance")

3: 动机与社会需求
("Strong need for Affiliation", "Indifferent to Affiliation")
("Strong need for Rejection", "Indifferent to Rejection")
("Strong need for Succorance", "Indifferent to Succorance")
("Strong need for Nurturance", "Indifferent to Nurturance")
("Strong need for Play", "Indifferent to Play")
("Strong need for Autonomy", "Indifferent to Autonomy")
("Strong need for Dominance", "Indifferent to Dominance")
("Strong need for Deference", "Indifferent to Deference")
("Strong need for Seclusion", "Indifferent to Seclusion")
("Strong need for Aggression", "Indifferent to Aggression")
("Strong need for personal achievement", "Indifferent to personal achievement")
("Strong need for recognition from others", "Indifferent to recognition from others")

4 沟通与人际互动风格
("Prefers superficial interaction", "Prefers deep interaction")
("Prefers direct communication to handle conflict", "Prefers avoidance, mediation, compromise")
("Concise communication style", "Detailed communication style")